

Regularized Model as a Candidate Baseline

Scientific Question

Is a simple regularized model a reasonable baseline before a larger experiment? This note uses synthetic, public-safe evidence to inform that bounded decision.

The running example is a prediction problem with paired observations (x_i, y_i) , where $x_i \in \mathbb{R}^p$ and y_i is a continuous endpoint. The baseline estimator solves

$$\hat{\theta}_0 = \operatorname{argmin}_{\theta \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n (y_i - x_i^\top \theta)^2.$$

The regularized candidate uses

$$\hat{\theta}_\lambda = \operatorname{argmin}_{\theta \in \mathbb{R}^p} \left[\frac{1}{n} \sum_{i=1}^n (y_i - x_i^\top \theta)^2 + \lambda \|\theta\|_2^2 \right].$$

The two estimators use the same squared-error term. The baseline minimizes that term alone, whereas the candidate adds a penalty on the squared coefficient norm. The parameter λ controls the contribution of the penalty to the objective. The comparison therefore asks whether this regularized candidate merits a place among the baselines for the larger experiment; it does not establish the best penalty strength or final method.

Evidence Design

The comparison is intentionally small. It is meant to check rendering and decision structure, not to claim a scientific discovery. We evaluate two synthetic folds, report root mean squared error, and inspect whether the regularized model is consistently lower without hiding the uncertainty.

Fold	Baseline RMSE	Regularized RMSE	Delta	Interpretation
A	1.42	1.18	-0.24	regularization helps
B	1.36	1.29	-0.07	effect is smaller
Mean	1.39	1.235	-0.155	promising but limited

Table entries compare the two models within each synthetic fold. Delta is the regularized RMSE minus the baseline RMSE, so a negative value favors the regularized candidate. The direction agrees across the two folds, while the magnitude differs. This agreement is the useful observation for the current decision, but it should not be read as evidence that the gain has a stable magnitude across future splits.

The average reduction is

$$\Delta_{\text{mean}} = \frac{1}{2} \sum_{k \in \{A, B\}} (\text{RMSE}_{\lambda, k} - \text{RMSE}_{0, k}) = -0.155.$$

The mean weights the two fold-level differences equally. It summarizes the observed comparison, but it does not describe a distribution of differences or quantify uncertainty. In particular, the smaller improve-

ment in fold B remains relevant even though the mean favors the regularized model. The fold-level values should remain visible alongside the aggregate when judging whether to carry this candidate forward.

Interpretation

The result supports using the regularized model as a conservative baseline in a larger experiment. It does not justify a claim that the regularized model is optimal, and it does not distinguish whether the improvement comes from variance reduction, implicit feature weighting, or fold-specific noise. The advisor decision is therefore narrow: keep the regularized model in the next comparison set, but do not make it the final method.

The main caveat is that two folds are not enough to estimate variability. A larger run should include at least five folds or repeated random splits, and the report should show the distribution of fold-level differences rather than only the mean. If the next run still shows a negative mean delta and no severe outlier fold, the method is worth using as the default baseline.